

Exact stability certificates for neural temporal knowledge graph completion

Abstract

Neural knowledge systems need to manage the evidence behind an answer when that evidence can be revised. Prediction confidence alone does not specify whether an answer survives the withdrawal of a coherent period of observations. We introduce StableKG, an editable temporal-memory interface with exact stability certificates for neural temporal knowledge graph completion. The interface combines a frozen neural distribution with a memory distribution that is renormalized after deletion. Multiplying a post-deletion pairwise margin by its retained-mass fraction reduces certification to an additive optimization, yielding an exact adversary and a constructive deletion witness. A sparse competitor theorem preserves the full-vocabulary guarantee, and a reverse index enables local maintenance. Experiments with two backbones and three training seeds on ICEWS14 and ICEWS05-15 show that certificate surplus improves deletion-flip discrimination over the original score margin by 0.064–0.584 AUROC across 16 perturbation comparisons. At 40% coverage, certified selection eliminates worst-window flips for the compact backbone and reduces the joint rate of incorrect or edit-sensitive answers by 11.07 and 4.05 percentage points on the two datasets. Across 240 edit transactions, including a 2.3-million-event GDELT history, local maintenance reduces query-state refreshes by 95.41%. These results establish an explicit, computable tolerance to evidence withdrawal as a complementary capability of neural graph prediction.